

Example 1.6

Find the acceleration of a proton moves in $E = 0.5 \text{ kN/C}$ directed to right

Proton: $q = 1.6 \times 10^{-19} \text{ C}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$

$$F = ma = qE$$

$$a = \frac{qE}{m} = \frac{1.6 \times 10^{-19} \times 500}{1.67 \times 10^{-27}} = 4.8 \times 10^{10} \text{ m/s}^2 \text{ (right)}$$

Example 1.7

Two opposite charge, the positive is two times the other.

if the electric field at the middle is 24 N/C . Find the new value of the

electric field if the two charges are brought in contact and the separate to the same distance.

Contact & separate:

q_1, q_2 , then

$$q_1 = q_2 = \frac{q_1 + q_2}{2}$$

$$E = k_e \frac{q_1}{r_1^2} + k_e \frac{q_2}{r_2^2}$$

$$24 = k_e \frac{2q}{(\frac{r}{2})^2} + k_e \frac{q}{(\frac{r}{2})^2}$$

$$24 = k_e \frac{8q}{r^2} + k_e \frac{4q}{r^2} \rightarrow 24 = 12 k_e \frac{q}{r^2}$$

$$k_e \frac{q}{r^2} = 2$$

Contact & separate:

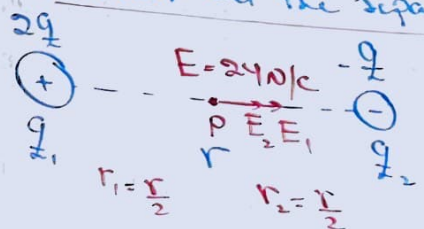
$$q_1 = q_2 = \frac{2q + (-q)}{2} = \frac{q}{2}$$

$$E_{\text{new}} = k_e \frac{q_1}{r_1^2} + k_e \frac{q_2}{r_2^2}$$

$$E_{\text{new}} = k_e \frac{q/2}{(r/2)^2} + k_e \frac{q/2}{(r/2)^2}$$

$$E_{\text{new}} = k_e \frac{2q}{r^2} + k_e \frac{2q}{r^2}$$

$$E_{\text{new}} = 4 k_e \frac{q}{r^2} \rightarrow E_{\text{new}} = 4(2) = 8 \text{ N/C}$$

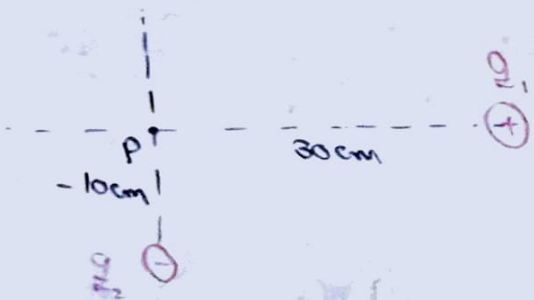


Example 1.8

Find the electric field at origin

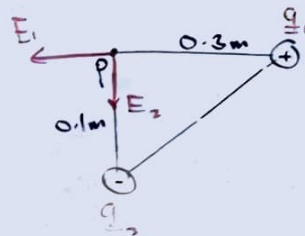
$$q_1 = 6 \text{ nC at } (30 \text{ cm}, 0)$$

$$q_2 = -3 \text{ nC at } (0, -10 \text{ cm})$$



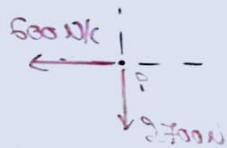
$$E_{2x} = -600$$

$$E_{2y} = -2700$$



$$E_1 = k_e \frac{q_1}{r_1^2} = 9 \times 10^9 \times \frac{6 \times 10^{-9}}{0.3^2} = 600 \text{ N/C}$$

$$E_2 = k_e \frac{q_2}{r_2^2} = 9 \times 10^9 \times \frac{3 \times 10^{-9}}{0.1^2} = 2700 \text{ N/C}$$



$$E_r = \sqrt{600^2 + 2700^2} = ?$$

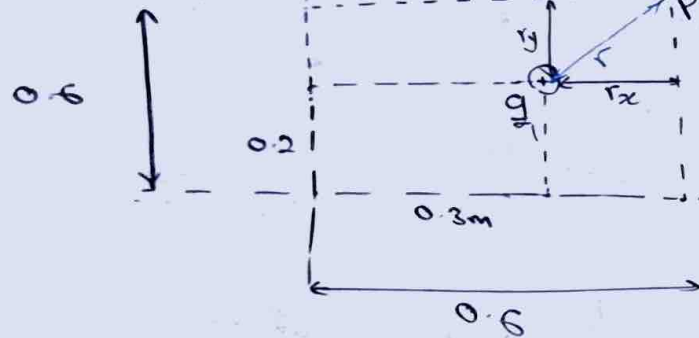
$$\theta = \tan^{-1}\left(\frac{2700}{600}\right) = ?$$

$$\theta_{tr} = 180 + \theta = ?$$

Example 1.8

$q_1 = 5 \mu\text{C}$ at (x_1, y_1)
 $(0.3\text{m}, 0.2\text{m})$

a) Find the electric field at point P which is placed at (x_2, y_2)
 $(0.6\text{m}, 0.6\text{m})$



$$\left. \begin{aligned} r_x &= 0.6 - 0.3 = 0.3 \\ r_y &= 0.6 - 0.2 = 0.4 \end{aligned} \right\} r = \sqrt{0.3^2 + 0.4^2} = 0.5$$

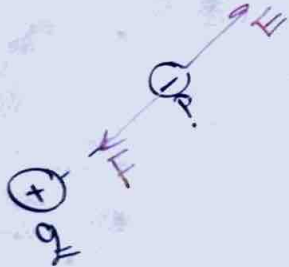
$$E = k \frac{q_1}{r^2} = 9 \times 10^9 \times \frac{5 \times 10^{-6}}{0.5^2} = 180 \times 10^3 \text{ N/C}$$

$$\theta = \tan^{-1} \left(\frac{r_y}{r_x} \right) = \tan^{-1} \left(\frac{0.4}{0.3} \right) = 53.12^\circ \text{ (with x)}$$

Example 1.8

b) Find the electric force on $-2 \mu\text{C}$ charge placed at point P.

$$F = qE = 2 \times 10^{-6} \times 180 \times 10^3 = \underline{0.36 \text{ N}} \quad \text{at } \underline{233.13^\circ}$$



$$53.13 + 180$$

← 53.13